

Beyond the Chatbot

Building LLMs into Business Systems



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What We'll Cover

01



Coding Agents

AI collaborators in the terminal

02



Structured Outputs

From conversation to contract — defining what LLMs deliver

03



Chain-of-Thought

Orchestrated reasoning

04



Program Recipes

Portable blueprints — when the prompt becomes the product

01

Coding Agents

AI collaborators that draft, debug, and iterate alongside you

Major AI Coding Agents Today

>_ Claude Code

Anthropic

Agentic coding in
your terminal.
Full codebase
access.

</> Codex (OpenAI)

OpenAI

Cloud-based.
Parallel tasks in
sandboxed
environments.

Claude Cowork

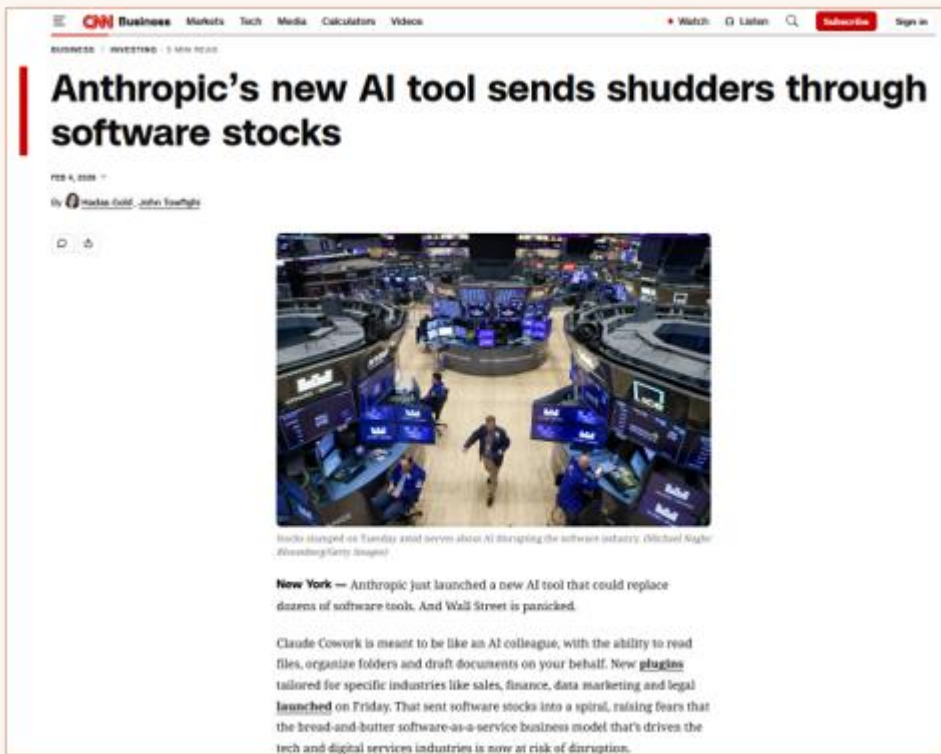
Anthropic (new)

GUI-based.
Automates file &
task
management.



Try many. Outages happen.

In the News



CNN · Feb 4, 2026

*AI Tool Triggers
Software Stock
Bloodbath*

[CNN Business, Feb. 4, 2026](#)

02

Structured Outputs

Teaching LLMs to deliver validated, typed data — not just text

The Problem: Unstructured Information

RAW ER INTAKE TRANSCRIPT

so yeah the patient is um a woman I think she said she's like 47 or maybe 48 and she came in saying her stomach has been really hurting for the past couple days like really bad cramping and she said she threw up twice this morning ... and she said the pain is mostly on her right side kind of low and it gets worse when she moves around or coughs and she has a slight fever the thermometer said 101 point something



How do you extract actionable data from this?

- ✗ No database storage
- ✗ No automated routing
- ✗ No urgency scoring

LLM Chat Can Clean It Up... Sort Of

RAW TRANSCRIPT

um ok so this guy came in hes I think 34 and hes saying his chest is really tight and hes been sweating a lot and also having trouble catching his breath and it started about two hours ago he said and he looks pretty anxious...



LLM CHATBOT

Male, approx. 34 y/o, self-ambulatory at arrival. Reports chest tightness, diaphoresis, and SOB onset ~2hr ago. Appears visibly anxious; skin moist/diaphoretic. No stated cardiac hx. Symptoms described as ongoing.



Now what?

Still text. Can't route, store, or automate.

Impressive conversion — but the output is still an opaque blob. Tough to act on it automatically.

Structured Outputs: A Simple Schema

Pydantic (Python)

```
class MedicalConversion(BaseModel):  
    medical_text: str = Field(  
        description="Rewrite the  
        intake transcript as  
        concise clinical notes  
        using standard medical  
        terminology"  
    )
```

JSON Schema (What the API Sees)

```
{  
  "title": "MedicalConversion",  
  "type": "object",  
  "properties": {  
    "medical_text": {  
      "type": "string",  
      "description":  
        "Rewrite..."  
    }  
  }  
}
```

Same output — but now **validated, typed, and inside a data structure**

Add One Field. Change **Everything**.

```
class MedicalConversion(BaseModel):  
    """Convert patient intake to  
    medical language."""  
    medical_text: str = Field(  
        description="...")  
    urgency_score: int = Field(  
        description="Rate 1-10 based on  
        the totality of the intake:  
        symptoms, urgency cues,  
        and what was not said"  
    )
```

A quantum leap in intelligence

One integer field. The model evaluates what was said, how it was said, and what was *not* said.

Traditional NLP misses context, nuance, and implication.

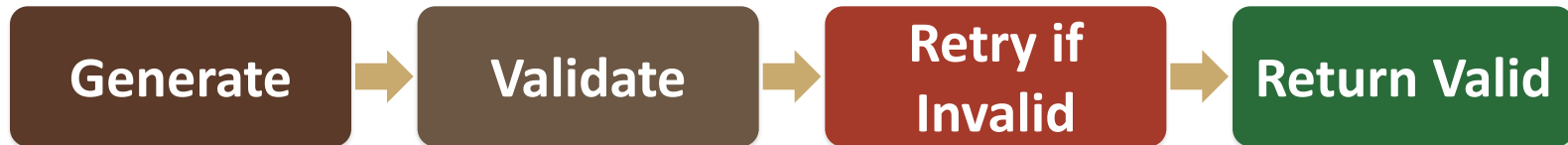
Under the Hood: How It Actually Works

PATH A: CONSTRAINED DECODING



Every token constrained at model level — output is **guaranteed valid**

PATH B: CODE ENFORCEMENT



Schema in prompt. **Retry with error feedback** until valid JSON.

Case Study: ER Triage



Triage

One word: what ER doc should do



RiskCalc

Worst case + time horizon?



SymptomCluster

Group symptoms: acute? pattern?



SymptomRead

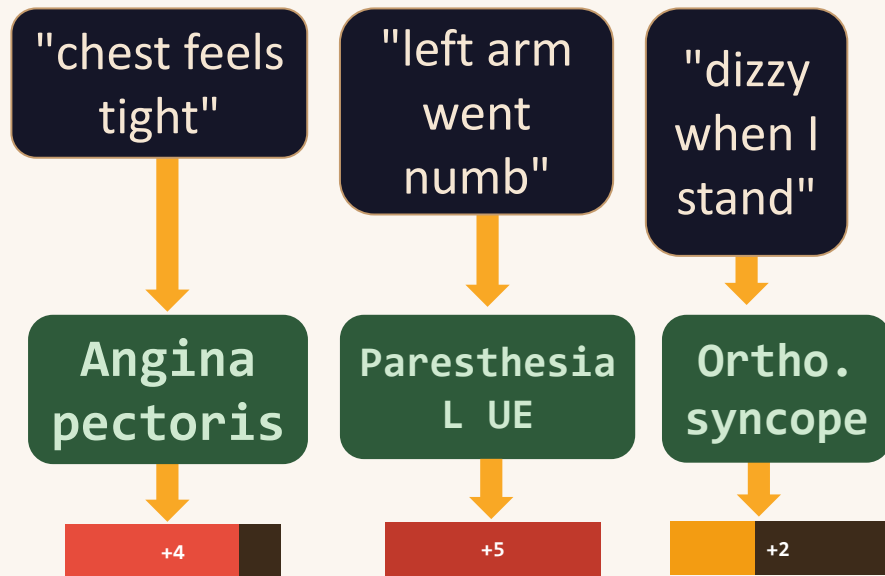
What did the patient say, and what does it mean?

Layer 1: SymptomRead

STEP 1 of 4

Patient language → structured clinical signal

```
class SymptomRead(BaseModel):  
    """Patient-reported →  
    clinical language."""  
    clinical: str = Field(  
        description="Medical  
        terminology.")  
    severity: int = Field(  
        description="-5=benign.  
        0=unclear. +5=acute.")
```



Each card = one **SymptomRead** object

Layer 2: ClinicalPicture

Pattern recognition across symptoms

STEP 2 of 4

```
class SymptomCluster(BaseModel):
```

```
    """Group related symptoms &
    flag acute clusters."""
```

```
    symptoms: list[SymptomRead]
```

```
    acute: bool = Field(
        description="True if
        requires immediate
        intervention. False
        if stable/non-urgent.")
```

Step 1

New



Cardiac Cluster

⚠ acute:
True

Angina
pectoris

+4

Paresthesia
L UE

+5

} symptoms cluster →

MI presentation

→ acute: True ⚡



Hemodynamic Cluster

✓ acute:
False

Ortho.
syncope

+2

Isolated, moderate

→ acute: False ✓

Layer 3: RiskCalc

STEP 3 of 4

Worst-case diagnosis + how fast the window is closing

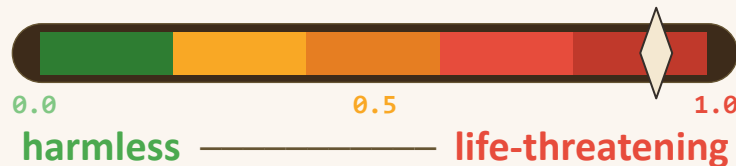
```
class RiskCalc(BaseModel):  
    """Score overall danger across  
    all symptom clusters."""
```

Step 2
clusters: list[SymptomCluster]

New
risk: float = Field(
 description="0.0=implausible
 or harmless. 1.0=near-certain
 and life-threatening. Weight
 plausibility × consequence.")



↓ ↙
risk = 0.92



● risk: 0.92

Near-certain & life-threatening

Layer 4: Triage

STEP 4 of 4

One word, carrying everything beneath it

```
TriageRouting = Literal[" surgery",  
"observation", "discharge", ...]
```

```
class Triage(BaseModel):
```

```
    """Final routing judgment."""
```

```
    calculations: list[RiskCalc]
```

```
    judgment: TriageRouting = Field(  
        description="One routing word  
        derived from the full chain:  
        symptoms → clusters → risks  
        → this final decision.")
```

Step 3

New

Angina

Pares.

Syncope



Cardiac



Hemodynamic



risk: 0.92 ← near-certain

judgment:
● "surgery"

→ Coronary Artery Bypass Graft

3 symptoms → 2 clusters → 0.92 risk → one word

Why Not Just Ask Directly?

Four reasons to chain structured steps instead of

Better thinking

Models reason better when you give them a staircase, not a cliff

Small models, big jobs

Break the problem down and suddenly a cheap model can do it

Show your work

Every intermediate step is a receipt you can audit later

You're the architect

Without structure, the model freelances its logic

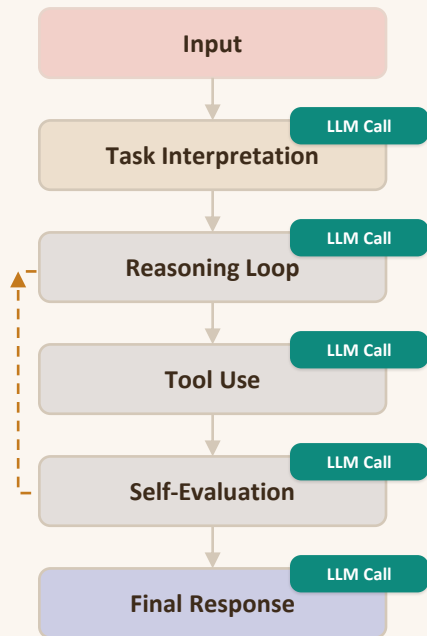
03

Chain-of-Thought in Practice

How structured outputs power reasoning

CoT + Action + Reflexion

An adaptive reasoning loop



[Yao et al., 2022 — ReAct](#)

KEY CONCEPTS

What is Chain of Thought?

- Organizes multiple stages.
- Passes each output as context into the next.

Why is it intelligent?

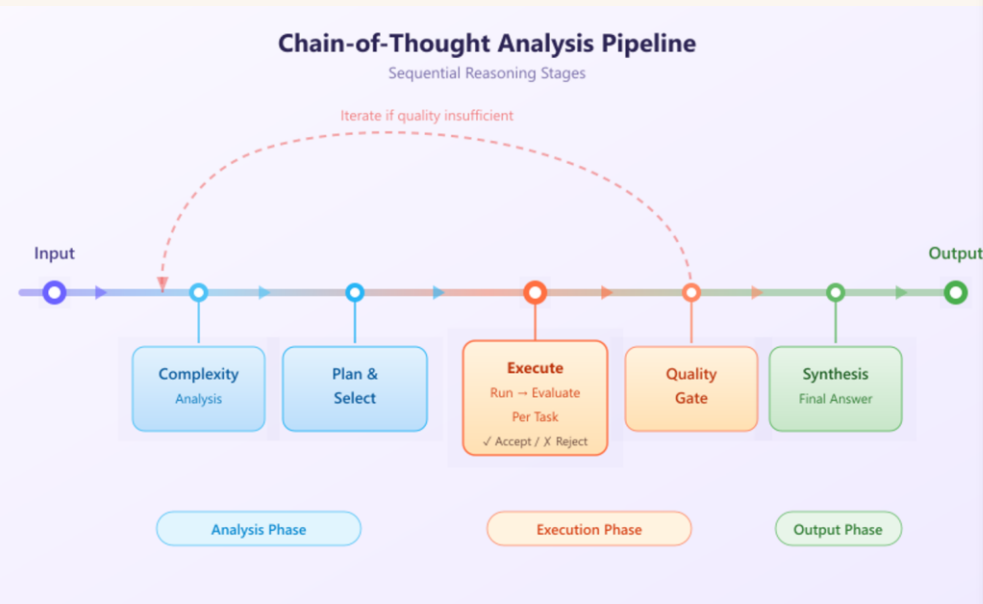
- Can change course based on prior results.
- Responds to what it discovers.

Structured Reasoning in Action



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— Live chain-of-thought analysis system



Small models, big reasoning

Structured steps let consumer-grade models produce analysis far beyond native capability.

Breaking Down the Problem

Structured outputs enable small models to tackle large problems

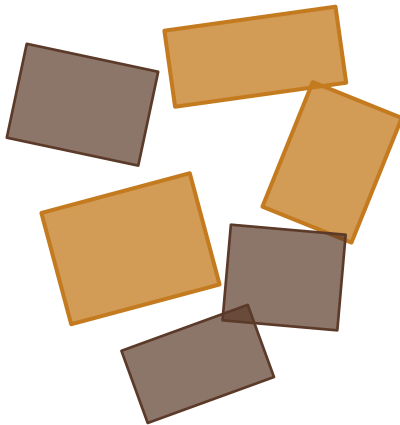
The Problem



*Too large for
a small model*



Naïve Split



*Pieces are random,
don't connect*



Structured Path



*Sequential path — each
piece builds on the last*

04

Program Recipes

If coding assistants are kitchens, prompts are recipes.

Inside the Recipe

When AI is the kitchen, the recipe is what counts

Prompt: Create a program called "ER Triage Structurer"

Step 1: Ingest ER Transcripts

- Read transcript files from `./data/er_transcripts/`
- Each file is one raw ER intake transcript.

Step 2: Extract with Structured Output

- Send each transcript to an LLM with structured output.
- Schema: **"Triage"**
 - └─ **calculations**: list of **«RiskCalc»**
 - └─ **clusters**: list of **«SymptomCluster»**
 - └─ **symptoms**: list of ... **«SymptomRead»**
 - └─ **clinical** (text) — medical term
 - └─ **severity** (int) — -5 to +5
 - └─ **acute** (bool) — immediate intervention?
 - └─ **risk** (decimal) — 0.0=harmless ... 1.0=critical
 - └─ **judgment**: one of [surgery | observation | discharge | ...]

Step 3: Validate and export structured JSON per patient

- Flag any transcript where `acute=true` or `risk ≥ 0.7`.

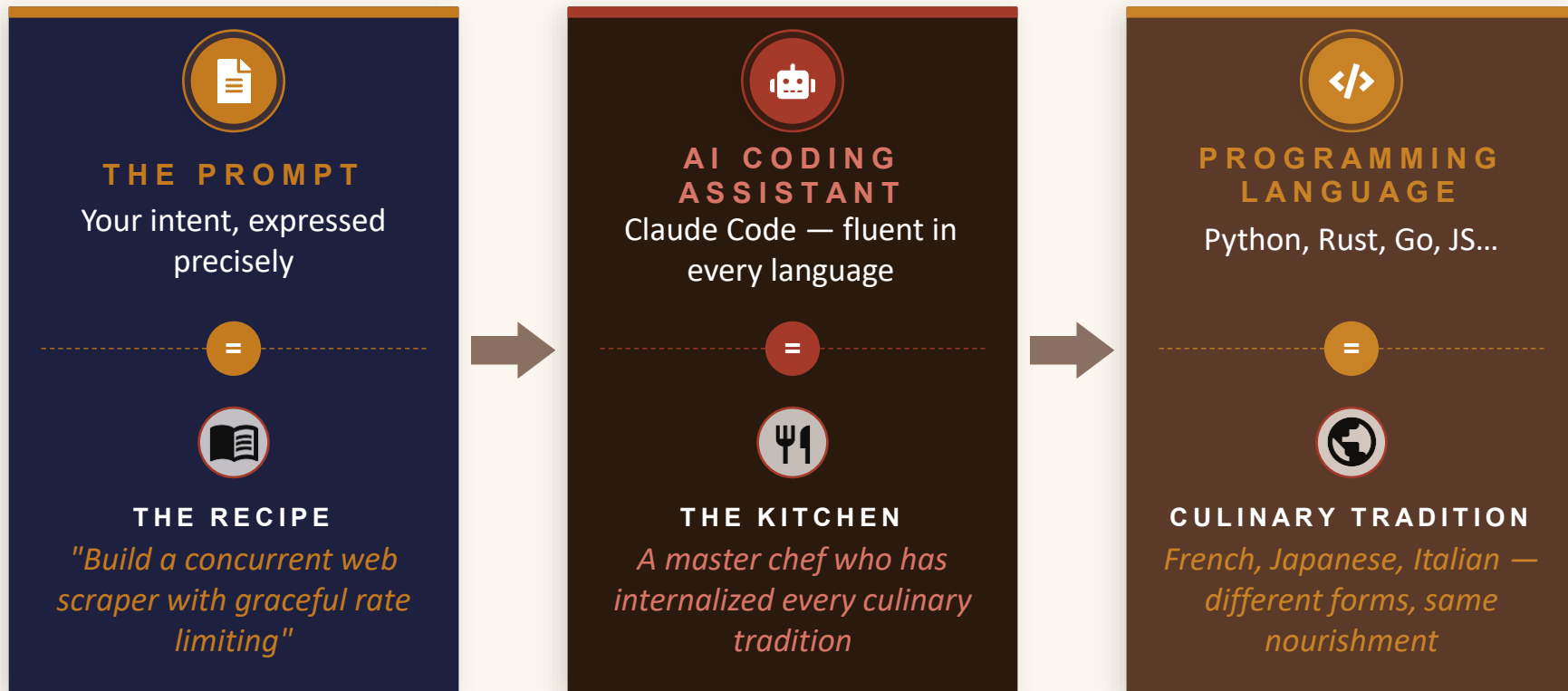
Apps by Instruction

Coding agents can turn prompts into full applications:

- Reads the spec you wrote
- Wires it into LLM API calls
- Builds the full application

The End of Language Loyalty?

When AI is the kitchen, the recipe is what counts





Thank You

Questions & Discussion

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